

EXPERIMENT I: GENERAL SURVEY OF PROKARYOTES, ALGAE,
FUNGI AND LOWER PLANT (CRYPTOGAMS)

ACTIVITY 1.1: MICROSCOPIC EXAMINATION OF FUNGI SLIDES

- Using the bread provided, pick a bit of the fungal growth with sharp needle.
- Place it on a clean slide
- Add lugol lactophenol blue
- Examine under microscope.
- Make diagram identifying the sporangium, sporangiophore and hyphae.

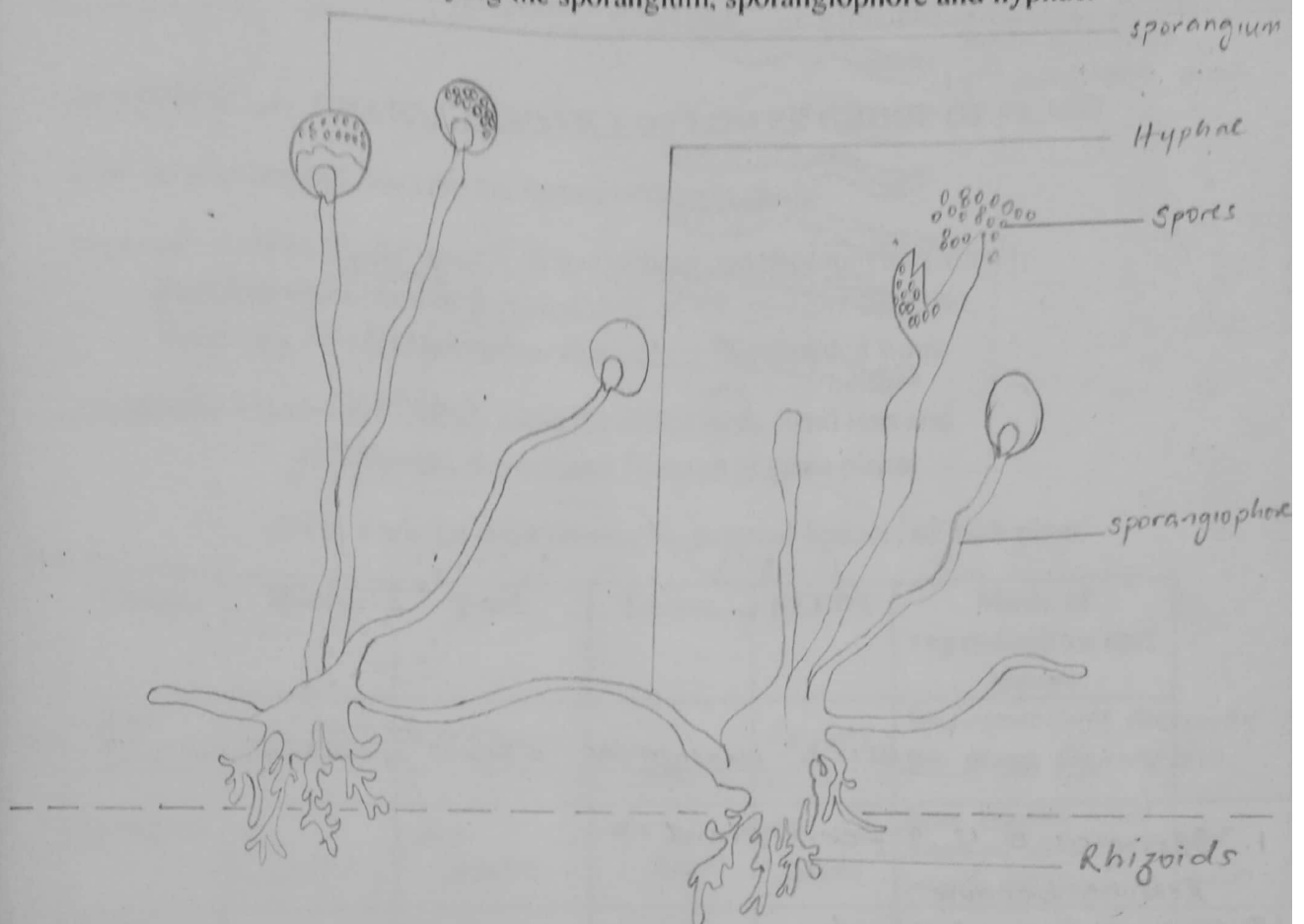
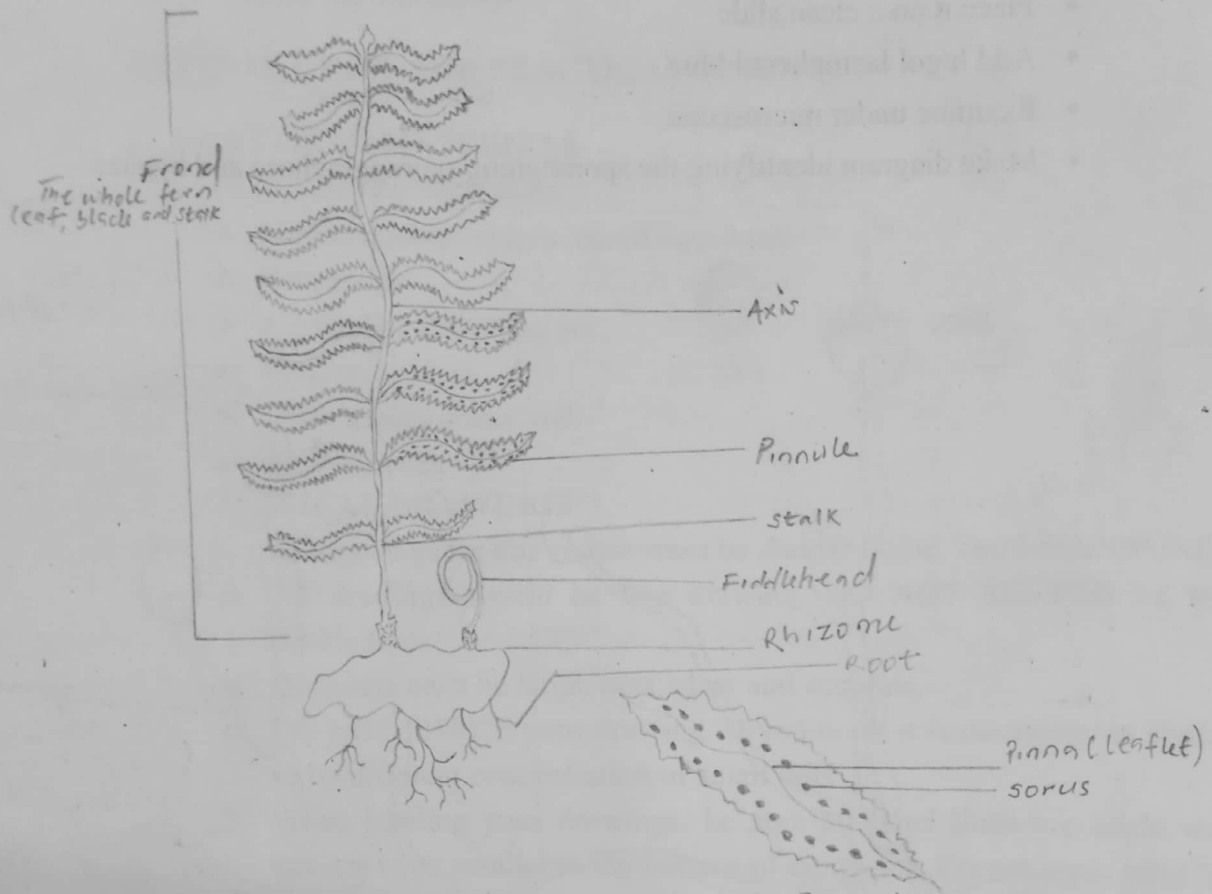


DIAGRAM OF A FUNGI ON A BREAD MOULD

ACTIVITY 1.2: LOWER PLANT (CRYPTOGAMS)

1. Identify specimen B provided with reasons.
2. Make a labeled diagram of a frond of specimen B showing the sori.
3. What is the significance of the sorus?

2.



A WELL LABELED DIAGRAM OF A FERN PLANT SHOWING THE SORI

1. Specimen B is a fern plant

Reasons because:

- i) The presence of fronds
- ii) Beneath the leaf you'll find the sorus

Presence of Fibrous root

3. What is the significance of the sorus

Sorus are spore producing structures their function is to release many spores.

ACTIVITY 1.3: FILL THE APPROPRIATE SPACE IN THE TABLE BELOW

	Examples	Characteristics
Prokaryotes	Bacteria, Blue-green algae and Mycoplasma	They have no nucleus and they lack organelle.
Algae	Chlorella, Brown algae, Red algae, Xanthophyta	All prokaryotic cells are enclosed by a cell wall. Algae are photosynthetic organisms. Algae are found where there is adequate moisture.
Fungi	Yeast, moulds, smuts, mushroom and puff balls	They reproduce by means of spores. They lack chlorophyll and hence can not perform photosynthesis.

13TH DECEMBER 2022

ACTIVITY 1.4: CHARACTERISTICS OF LOWER GROUP OF PLANT

Aim: to observe and describe the features of lower plants

Materials: Fungus (Bread mould), Algae (Spirogyra), Bryophytes (Moss), pteridophytes (fern or prepared slide of *Selaginella*), Dropper, Hand lens, Microscope, Glass slide, cover slip distilled water

Procedures: (1) observe closely using the naked eyes, hand lens and microscope, the external features of these plants.

(2) In a tabular form present the external features of each plant.

Plants	Roots	Stem	Leaves	SEEDS	Mode of reproduction and means
Bread mould	Hyphae	sporangia	no leaves	Spores	They reproduce asexually by spore formation
Spirogyra	No roots	No stem	No leaves	No seeds	Both sexual and asexual or vegetative mode of reproduction
Moss	Rhizoids	Phyllids	Phyllids	Spores	Vegetative and sexual mode of reproduction.
Fern	Rhizomes	Rhizomes	fronds	Spores	They reproduce by spores - also known as vegetative cloning
Selaginella	Meristems	protoplasts	Ligulate leaves	Seedless Vascular plant	Vegetative as well as sexual reproduction - eg fragmentation budding.

OBSERVATION

EXTERNAL FEATURES

- ① Bread Mould
Presence of hyphae
and spirogyra

sporangium
sporangiophore
stolon
hyphae
rhizoids

- ② Spirogyra
Visible as a slimy green mats
Filamentous masses come to the
surface and become visible
as slimy green mats

Presence of cell wall
nucleus
Pyrenoid
Spiral chloroplast

- ③ Moss
They have very tiny leaves the different
leaf shape are a useful clue to
identify which moss you are observing

They have capsule
sporophyte
thread like roots called rhizoids
stems are weak and free standing

- ④ FERN
Presence of root stems and leaves
Presence of frond

Presence of Rhizome
Presence of sori
Presence of pinna

- ⑤ Selaginella
They are creeping or ascendant plant
with simple scale like leaves {microphylls}
on branching stem from which roots also
arise

Presence of stem
cone
Rhizophore
leaf
root

QUESTION: Differentiate between Spirogyra and Bread Mould Compare and contrast between Moss and Fern?

Differentiate between Spirogyra and Bread mould

<u>Spirogyra</u>	<u>Bread Mould</u>
① They are Autotrophs i.e. They produce their own food	They are saprophyte, they obtain their own food from dead decaying matter.
② They belong to the kingdom plantae	They belong to Kingdom Fungi
③ They possess chlorophyll	Absence of chlorophyll
④ Their cell wall are made of cellulose	Their cell wall is made of chitin
⑤ They store their food as starch	They store their food as glycogen.

COMPARE AND CONTRAST BETWEEN MOSS AND FERN
Similarities between Fern and moss

- a) They are both primitive or early plant.
- b) Both mosses and fern shows alternation of generations
- c) They are non flowering plant and seedless
- d) They both reproduce Asexually
- e) They both produce spores

Difference between Fern and Moss

<u>FERN</u>	<u>MOSS</u>
① Large spore producing sori	Small spore production
② Presence of roots and stem	Absence of root and stem
③ Presence of true leaves	Presence of leaf like structure
④ Belong to the phylum pteridophyta	Belong to the phylum Bryophytes
⑤ Vascular plant	non Vascular plant

13th DECEMBER 2022
EXPERIMENT II: GENERAL SURVEY OF HIGHER PLANTS AND ANIMAL KINGDOM

ACTIVITY 2.1: HIGHER PLANTS

You are provided with specimen A-E. Study them carefully and answer the questions below.

1. Identify specimen A without reasons.
2. To what group plants does specimen A belong?
3. Draw specimen B showing all the features you can see.
4. In two ways, how does specimen B differ from A considering the group of the plant to which B belong?

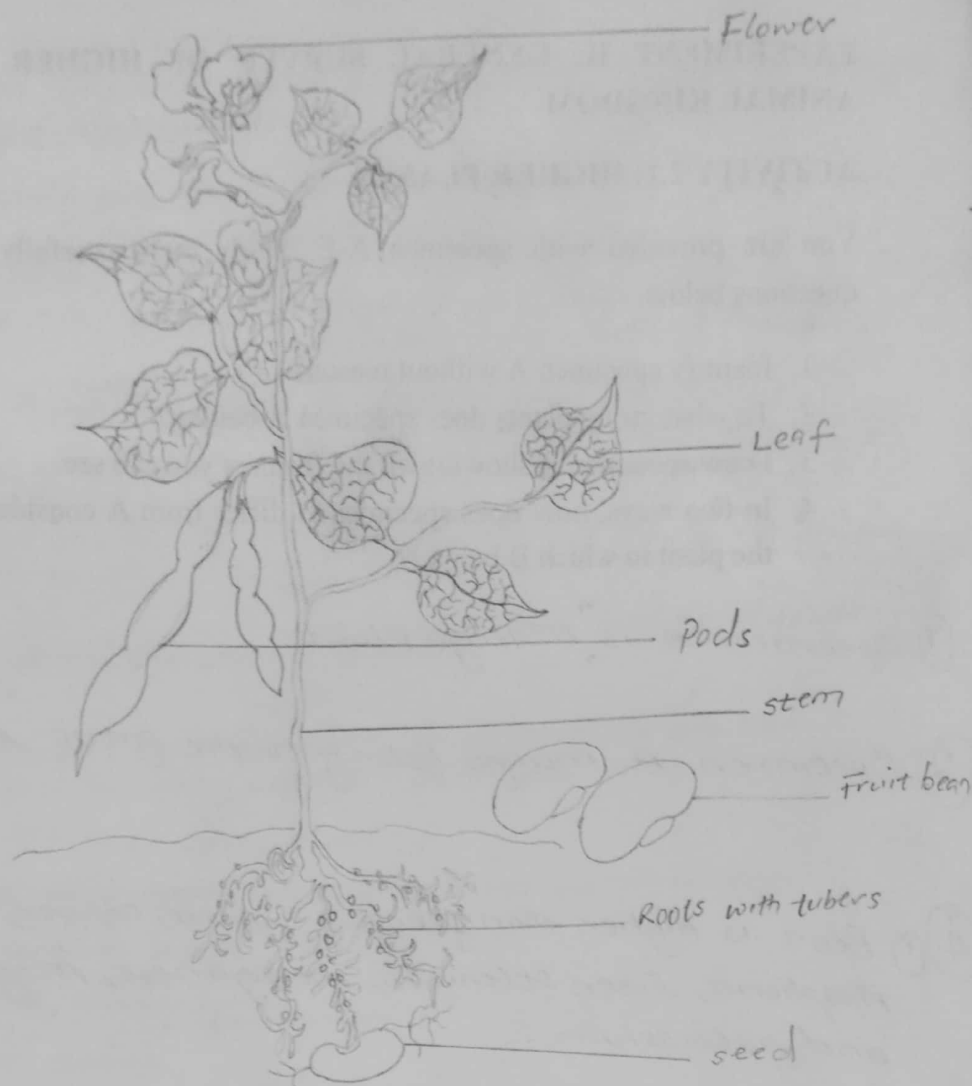
① Specimen A is a mango plant

② Specimen A mango belong to the group ANACARDIACEAE

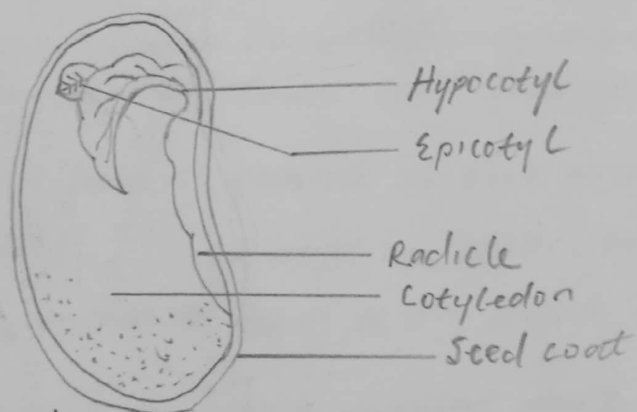
④ ① Bean is higher than mango in folate, copper, iron, Vitamin B₁, phosphorus, fiber, selenium, manganese, magnesium — and potassium —

①① Bean cover daily need of folate 121% more than mango

B.



THE DIAGRAM OF A BEAN PLANT



Bean seed Dicot

ventral nerve cord then extend along the floor of the body cavity to the last segment).

9. Find the excretory system:
(i.e. the nephridium which consists of a pair of minute, coiled, white tubules located in every segment except the first three and the last. Each nephridium opens to the outside by means of an excretory pore).
10. Identify the reproductive system:
(Check segments 9 and you will find SEMINAL RECEPTACLES and SEMINAL VESICLES: the seminal receptacles are small, white special bodies and seminal vesicles are light coloured bodies- through segment 12 that house maturing sperm that have been formed in two pairs of testes within them and sperm ducts that pass opening in segment 15).

OBSERVATION/RESULTS:

In the observation the following organs were observed, pharynx, heart, dorsal blood vessel, crop, gizzard, intestine, reproductive organs — and ventral cord

ANY OTHER COMMENTS:

Just like a human, an earthworm consumes food through one end and excrete waste through another, extracting the nutrients it needs as the food passes through the digestive tracts.

CONCLUSIONS:

REVIEW QUESTIONS:

1. Does the digestive system show specialization of parts? Explain.
2. Does the excretory system show that the earthworm is segmented? Explain.
3. Does the earthworm have an open or closed circulatory system? Explain.
4. Is the earthworm hermaphroditic? Explain.
5. Does the earthworm have a respiratory system?
6. How does it exchange gases?
7. Why would you expect an earthworm to lack an exoskeleton?

① Yes

Earthworm have a one way digestive system with a mouth and an Anus. This result in specialization of the different parts of the digestive tract in earthworm, this is the mouth, pharynx, Oesophagus, crop {for storing food} Gizzard {grinds food}, intestine {which contain a large dorsal-fold - typhlosole} to provide increase for digestion and absorption and anus. Soluble food passes through the walls of the small intestine into the blood.

② Yes

Nephridium is a coiled tubules that regulate the volume and composition of the body fluid and thus act as the excretory organ in earthworm. Nephridia are arranged in three segmentary {3 - last segment} and pharyngeal nephridia {4 - 6 segment}.

③ Earthworm has a closed circulatory system.

Earthworm circulates blood exclusively through vessels. There are three main vessels that supply the blood to organs within the earthworm - they are aortic arches, dorsal blood vessel and ventral blood vessel.

④ Yes

Earthworm are hermaphrodites where each earthworm contain both male and female sex organ can produce both male and female sex gamete {Egg & sperm}, although they are hermaphrodites, they need to mate.

⑤ Earthworm do not have respiratory system
Earthworm Breathe through their skin by diffusion. For diffusion to take place the earth skin must be kept moist.

⑥ Earthworm exchanges gas through their moist skin.
The exchange of gases take place passively through their skin.

⑦ Earthworm lack exoskeleton because their bodies are supported by hydrostatic skeleton. This means that their bodies are structurally supported and their muscle attachment rely on a coelom, a series of fluid-filled chamber that use pressure rather than bones or an external covering.

ACTIVITY 2.3 PHYLUM ARTHROPODA

13TH DECEMBER 2022

Observe specimen ^DF provided.

Identify and classify ^DF into kingdom, phylum and class.

Mention any 2 characteristic features of specimen ^DF.

Specimen ^D is a flattened oval body, long thread like antennae and a shining black or brown leathery integument.

Specimen ^D is a cockroach because it is wide, flat, oval shaped body.

Cockroach belong to the kingdom ANIMALIA

Cockroach belong to the Phylum ARTHROPODA

Cockroach belong to the CLASS INSECTA

Characteristic features of cockroach

- i) They ~~are~~ ^{have} flattened ~~oblong~~ oval body.
- ii) They undergo incomplete metamorphosis :
Egg → nymph {incomplete adult} →

Characteristic features of cockroaches

- i) They have flattened oval body
- ii) Long threadlike antennae and a shining black or brown - leathery integument.

What is the function of the TYMPANUM?

What is the function of the NARES?

What is a frog's method of locomotion on land and in water?

What is the function of the GLOTTIS?

Explain the term UROGENITAL SYSTEM?

The CLOACA receives materials from Intestine, urinary and Reproductive system.

(a) The function of the tympanium is to allow the toad to hear both in air and below the water. It receive sound wave and transmit them to the inner part of the amphibian ear.

(b) The function of the Nares is to act as a valve controlling the flow of air between the buccal cavity and the atmosphere and the glottal valve controls the flow of air between the buccal cavity and lungs.

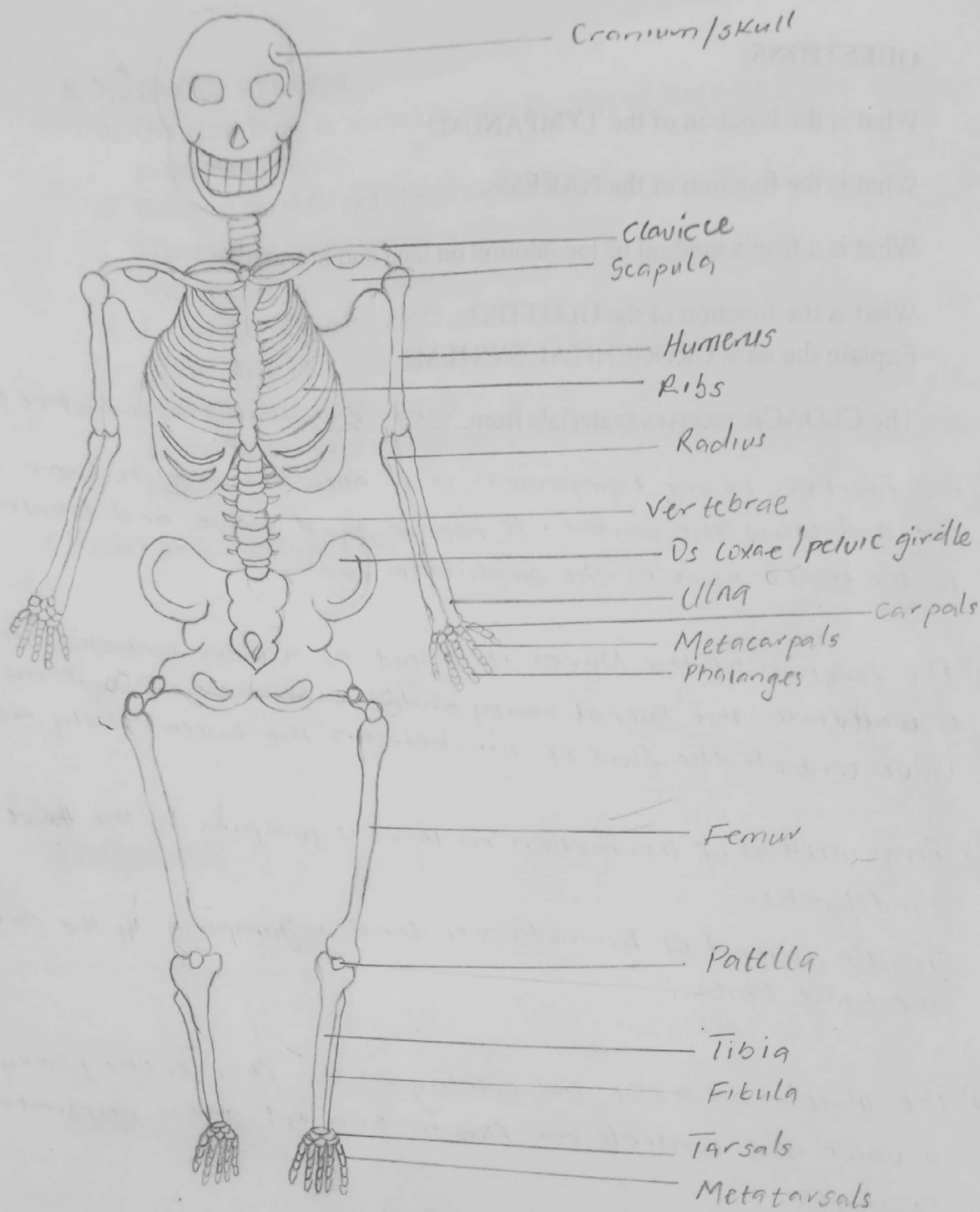
(c) Frog's method of locomotion on land is jumping by the thrust of the hind limbs.

Toad's method of locomotion on land is swimming by the thrust of the hind limbs.

(d) The glottis are a slit like opening on the floor of the pharynx is a valve that controls air flow in and out of the respiratory passages.

(e) The Urinary and Reproductive system are closely associated in toad and are jointly referred to as UROGENITAL SYSTEM. It is primarily involved in assisting the toad in carrying out urine excretion and reproduction.

(f) The cloaca receives material from the INTESTINE, THE URINARY SYSTEM and the REPRODUCTIVE SYSTEM.



A WELL LABELED DIAGRAM OF A HUMAN SKELETON

ACTIVITY 2.5 PHYLUM CHORDATA F

Identify specimen & F

Draw a well labeled diagram of V. F

Give one function of each labeled parts of specimen & F

Specimen F is the Human skeleton.

Functions of the labeled part of a Human skeleton.

(i) Cranium/Skull: protects the brain.

(ii) Mandible: It holds the lower teeth in place, it assist in mastication, and forms lower jaw line.

(iii) Humerus: It serves as an attachment to 13 muscles which contribute to the movements of the hand and elbow.

(iv) Clavicle: It serves as a connection between the axial and appendicular skeleton in conjunction with scapula.

(v) Radius: permits the forearm and hand to pronate and supinate, flex and extend at the elbow.

(vi) Ribs: The ribs forms the main structure of the thoracic cage protecting the thoracic organs. However their main function is to aid in respiration.

(vii) Vertebrae (C, L, S, C): The vertebral column protects the spinal cord.

(viii) Os coxae or pelvic girdle: to connect and support the thighs to each side of the trunk of the body.

(ix) Ulna: it assists in pronation and supination of the forearm and hand.

(x) Carpals, meta carpals: Carpals helps to dissipate axial loading during weight bearing. While

Metacarpals: accommodates the muscles, tendons and nerves of the palm.

(xi) Phalanges: of both the finger and foot helps us to manipulate our environment, the foot helps us balance.

(xii) Femur: holds the weight of the body when standing and moving.

(xiii) Patella: to improve the effective extension capacity of the quadriceps muscle by increasing the moment arm of the patellar ligament. Protecting knee joint.

(xiv) Tibia: main function tibia is weight-bearing.

(xv) Fibula: to combine with the tibia and provide stability to the ankle joint.

(xvi) Tarsals / metatarsals: combination of the both bones forms longitudinal arch in the foot - a shape well adapted for carrying and transferring weight in bipedal locomotion.

Specimen A1 {Leaf plucked outside}

Experiment	Observation	Inference
Specimen A ₁ + Boiling water	The cells and enzymes becomes inactive and starch grains present was bursted.	The cells weakened and died.
Specimen A ₁ + 70% of alcohol {ethanol}	The leaf is decolourized	The leaf decolourize to test the presence of starch
Specimen A ₁ + boiled water in a beaker	The leaf was softened	
Specimen A ₁ on a white tile + few drops of iodine solution.	The leaf turns blue-black after iodine solution was added	The presence of starch is confirmed.

Specimen A2 {Leaf kept in a cupboard}

Experiment	Observation	Inference
Specimen A ₂ + Boiling water	The cells were softened and killed	The cells died
Specimen A ₂ + 70% of alcohol {ethanol}	The leaf is decolourized	The leaf decolourized to test for the presence of starch.
Specimen A ₂ + boiled water in a beaker	The leaf is softened	
Specimen A ₂ on a white tile + few drops of iodine solution	The leaf remains colourless there was no colour change	There is no presence of starch in the leaf.

ACTIVITY 3.2 TEST FOR CLASS OF FOOD SPECIMEN B

- Make a drop of B on a filter paper by crushing it
- Observe the spot against a source of light
- State the observation.
- From your observation which class of food is this [B].
- Mention 5 other food substances that belong to this class of foods

ANSWER

Experiment	Observation	Inference
A drop of specimen B on a filter paper and observe the spot against a source of light	The spot specimen B was dropped and smeared on becomes translucent.	This shows the presence of fats and oil. {lipids?}

* Specimen B belongs to the class of food of fats and oil

* List of food that belongs to fat and oil

- ① Fish
- ② Cheese
- ③ Groundnut.
- ④ Butter
- ⑤ Lard

ACTIVITY 3.3 TEST FOR CLASS OF FOOD SPECIMEN C

- Take a small quantity of specimen C solution in a test tube.
- Add 1cm of sodium hydroxide and 1% copper [ii] solution in drops.
- Shake the mixture after each drop
- State the observation
- From your observation which class of food is this [C].
- Mention five(5) other food substances that contain specimen C

Experiment	Answer Observation	Inference
Specimen C in a test tube + 1cm of sodium hydroxide solution	No visible reaction	
Specimen C + $\text{NaOH}_{(aq)}$ + 1% of copper II solution and shake	mixture turns to purple colour	Presence of protein is confirmed.

* From the observation, specimen C belongs to the class of food — called protein.

* Foods that contain specimen C are:

- ① Beef
- ② Egg
- ③ Milk
- ④ Soya bean
- ⑤ Chicken

EXPERIMENT IV: DIVERSITY AND CHARACTERISTICS OF PLANTS AND ANIMALS.

ACTIVITY 4.1: DIVERSITY AND CHARACTERISTICS OF PLANTS

You are provided with two specimens labeled A and B.

- i. Identify specimen A and B.

A: Fern plant

B: Mango plant

- ii. What class of plant does specimens A and B belong to?

A: Class: Polypodiopsida

B: Class: Magnoliopsida

- iii. Identify the similarities in their physical features

1. Similar to Mango plant, ferns have roots, stems and leaves.
2. Ferns possess curled stems {called "fiddleheads"} that open into leaves {called "fronds"}. While Mango possess a long, string-like stem.
3. Both ferns plant and mango plant possess or contain chlorophyll. It is green in colour of leaves also - Bicolour.

iv. Identify the difference in their physical features

FERN	MANGO
- Fern plant are non flowering plant and they do not produce fruits	- Mango plant are flowering plant and they produce fruits.
- Ferns are a hardy perennial plant that will grow back each year.	- Mango plant are purely perennial, they live well past 100 years and can produce fruit until the late stages of their life cycle.
- Ferns leaves are pinnate in shape	- Mango leaves are ovate in shape shape
- Ferns leaves {frond} are between 18-24 inches long	- Mango leaves are between 5 to 16 inches in length.

v. Mention any two external features of specimens ^{A B} C and D and state their function(s).

Specimen	Feature	Functions
A	Fronds	Responsible for reproduction and for photosynthesis
	Sporangia	They help in making and storing spores.
B	stem	It is responsible for carrying water and mineral salt absorbed by the root
	Root	It is responsible for the absorption of water and mineral salt.

ACTIVITY 4.2: DIVERSITY AND CHARACTERISTICS OF ANIMALS

You are provided with two specimens labeled C and D

i. Identify specimens C and D

C: *Ascaris lumbricoides*

D: Pigeon

ii. What class of ~~plant~~^{animal} does specimens C and D belong to?

C: Chordata

D: Aves

iii. Identify the similarities in their physical features

- ① They both possess mouth and beak^{respectively} for ingestion
- ② They both possess excretory pores and anus for excretion

iv. Identify the differences in their physical features

<i>Ascaris</i>	Pigeon
- <i>Ascaris</i> are microscopic organisms	- Pigeon are macroscopic organisms.
- They have no limb	- They possess 2 limbs
- They have no wing	- They possess a pair of wings
- They have no feather - hence they are transparent due to presence of cuticle and hypodermis in their skin	- Body is covered in feathers

v. Mention any two external features of specimens C and D and State their function(s).

Specimen	Feature	Function
C	Cuticle and hypodermis	Protects the skin from the acids in the intestine.
	Excretory pore	An opening on the body surface for excretion.
D	Claws	They are used for walking and perching.
	Beaks	They are used for ingestion, fighting and preening of feathers.

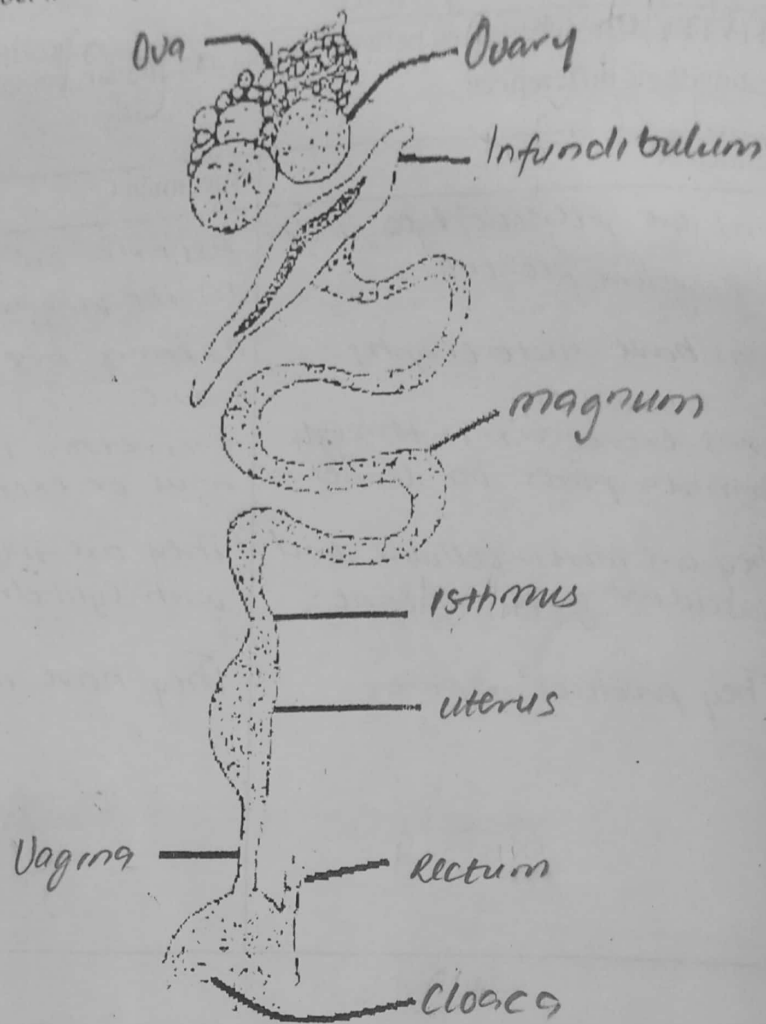
ACTIVITY 4.3: differences between plants and animals compare specimen A and C, noting their differences

Specimen A	Specimen C
1) Ferns are pteridophyte under the Kingdom plantae 2) Ferns are autotrophs 3) Ferns excretion is through stomata pores on leaves 4) They are multi-cellular and vascular green plant. 5) They produce spores	Ascaris are found in the Kingdom Animalia. Ascaris are parasitic in nature Excretion is through the anus or excretory pore They are microscopic organism with cylindrical bodies. They have no distinct head

EXPERIMENT V: GENERAL REPRODUCTION AND
INTERRELATIONSHIP OF ORGANISMS

ACTIVITY 5.1: SEXUAL REPRODUCTION IN ANIMALS

The diagram below represents the female reproductive system of a fowl. Supply the correct label in the spaces provided:



- i. Which of the structures are eggs formed?

The eggs are formed in the

- ii. Egg shells are formed in

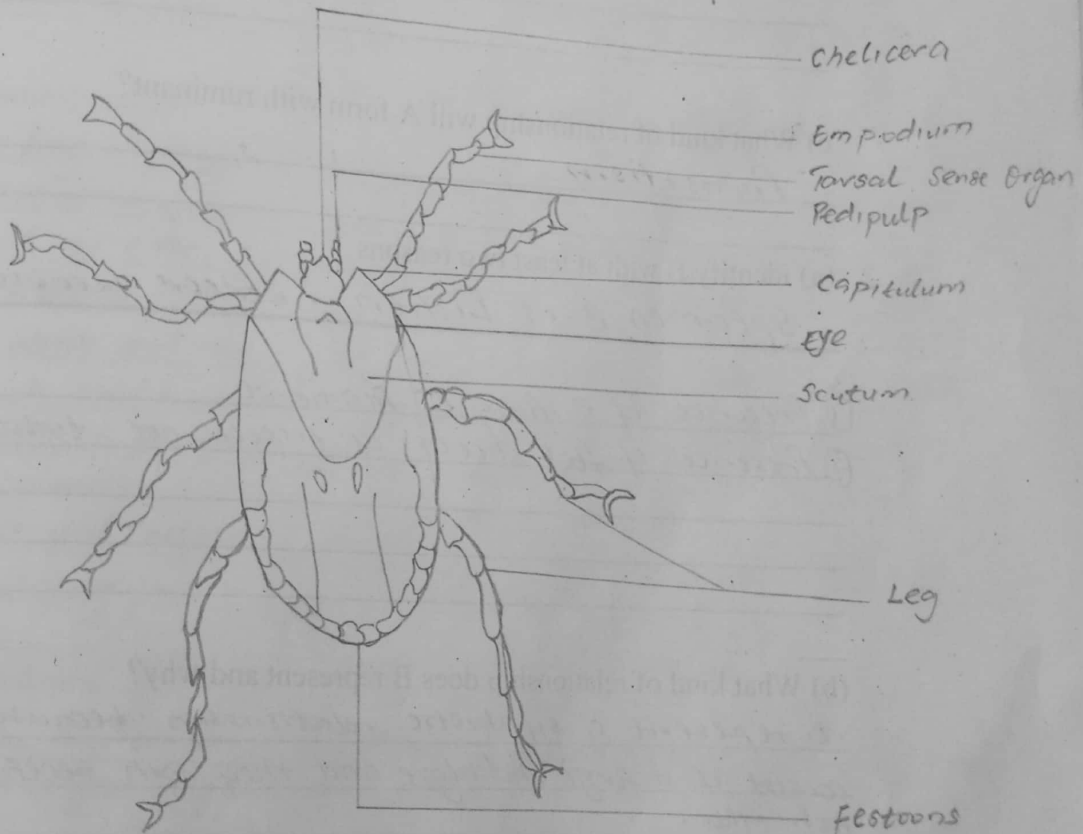
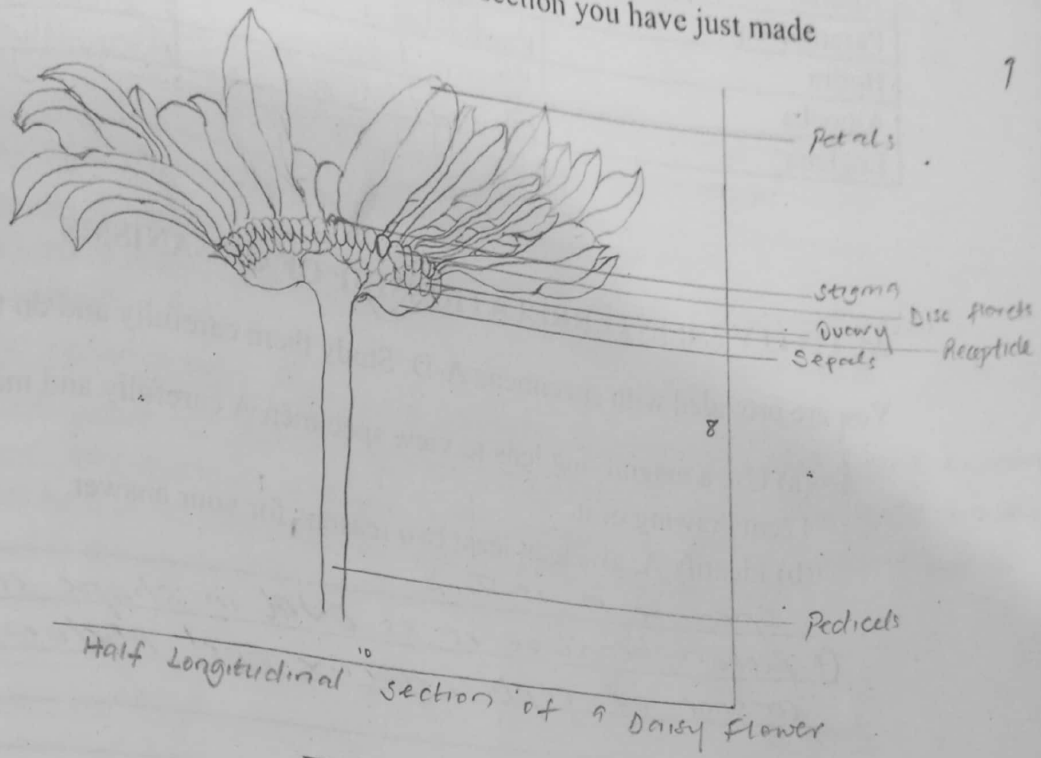
Uterus

- iii. A complete egg is laid through the

A Complete egg is laid through the Cloaca

You are provided with the following structure of a plant.

- i. Carefully make a longitudinal section of the specimen
- ii. Draw and label a half of the longitudinal section you have just made



ACTIVITY 5.4

39

LABELLED DIAGRAM OF A TICK. {SPECIMEN A}

ACTIVITY 5.3: ASEXUAL REPRODUCTION IN ANIMALS

Complete the table below:

Animal	Mode of reproduction	Type of reproduction
Paramecium	Binary fission	Asexual reproduction
Hydra	Budding	Asexual reproduction
Amoeba	Binary fission	Asexual reproduction
Euglena	Binary fission	Asexual reproduction

ACTIVITY 5.4: INTERRELATIONSHIP OF ORGANISMS

You are provided with specimens A-D. Study them carefully and do the following.

1. (a) Use a magnifying lens to view specimen A carefully and make a 10-12cm drawing of it.

- (b) Identify A, giving at least two reasons for your answer.

Specimen A is Tick

- ① Reason because it is oval in shape and it consist of cephalothorax and Abdomen

- c) What kind of relationship will A form with ruminant?

Parasitism

2. (a) identify B with at least two reasons

Specimen B is Lichen - Reason because

- ① Presence of a fungal filament.

- ② Presence of flat sheets of tissue not tightly bonded.

- (b) What kind of relationship does B represent and why?

B represent a symbiotic relationship because a lichen consist of a fungi and algae and they both benefit from each other.

(c). mention the components of B

A fungus and a chlorophyll containing algae or -
cyano bacteria.

3. (a) Identify specimens C and D.

C:

Specimen C is Ascaris lumbricoides

D: Specimen D is Cowpea

(b) What kind of relationship would C form with man and why?

Parasitism, because the host is affected while it (Ascaris) gains.

(c). what kind of relationship is represented by D? What is the plant's role in the relationship?

Symbiotic relationship. The plant enables the nitrogen fixing soil bacteria called rhizobia to form nodules on the plant roots, allowing the bacteria to convert atmospheric nitrogen into ammonia that the plant use.

4. (a). For each of the following, name one example that has not been mentioned in the course of this practical session and state why:

(i). symbiosis:

① Hermit crabs and Sea Anemones: Sea Anemones eat scrap of food that the hermit crab releases, the hermit crab is protected from predators.

(ii). Commensalism:

① Tree frogs on plant: Tree frogs use large leafy plant and trees to protect themselves from rain. This does not affect the plant or trees, but it help the frogs stay safe.

(iii). Parasitism:

Tape worm and man: The tapeworm attach itself to the host stomach and sucks nutrient from the host thereby depriving the host of nutrients.

(iv). Competition:

Lions and cheetahs: compete for the same antelope and gazelle prey.

(v). predation:

Hawks and chicks: the hawks hunt and chase the chick until it gets it and kills it.

EXPERIMENT VI: HEREDITY AND EVOLUTION.

ACTIVITY 6.1: UNDERSTANDING HEREDITY (CHROMOSOMES)

Aim: To prepare and examine chromosomal slide structures.

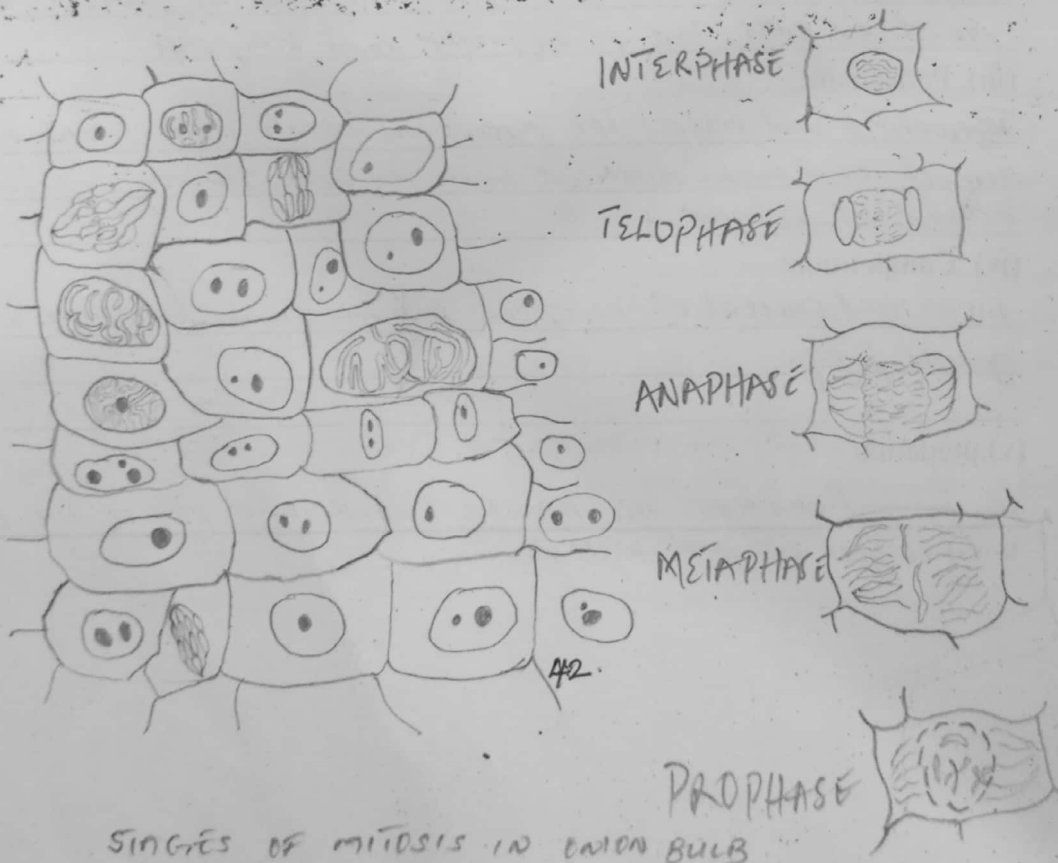
Method:

- Cut thin sections of 2-5mm from the root tips of the onion bulb provided.
- Place each section on a microscopic slide.
- Add a drop of aceto-carmine solution.
- Tease out and squash the material gently with a needle.
- Remove all large pieces of debris.
- Cover with a cover slip and heat gently over a Bunsen flame for a few seconds.

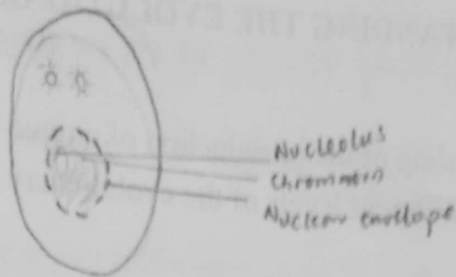
EXERCISE:

Examine the preparation carefully under the microscope and observe the different stages of mitosis. Make a drawing of the structures observed

DRAW IN THE SPACE PROVIDED BELOW:



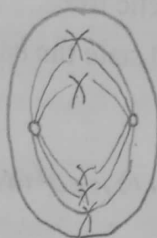
x50



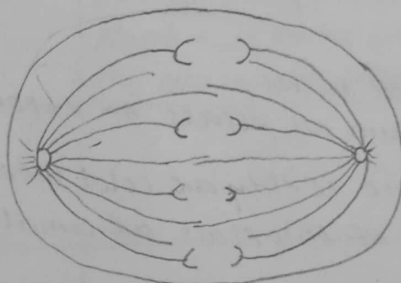
Interphase $\times 3$



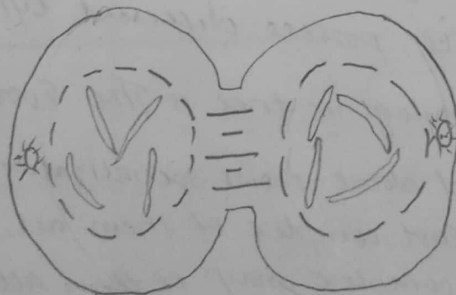
Prophase $\times 3$



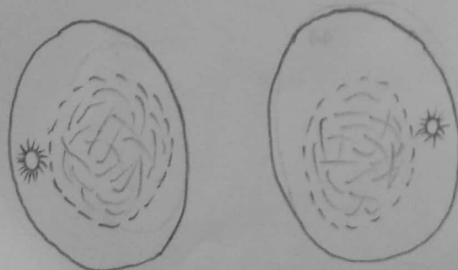
Metaphase $\times 3$



Anaphase $\times 3$



Telophase $\times 3$



Cytokinesis $\times 3$

ACTIVITY 6.2: UNDERSTANDING THE EVOLUTIONARY RELATIONSHIPS

AIM: To establish understanding of the introduction of unique features distinguishing organisms at different levels of the evolutionary tree.

METHOD:

You are provided with three different projected images of the phylogenetic tree with a brief narration. Study the last Phylogenetic tree, it illustrates the different hypothesis in functions of systems and the existing relationship (attempts the exercise).

EXERCISE:

- List 3 examples of animals on the 3 selected branches of the tree
- State 2 characteristics common to each members on branches of the tree
- Give an alternative name of the phylogenetic tree
- Examine the complexity of organisms and state the basic relationship existing from bottom to top.

Answers

① Arthropods: cockroaches, spiders, millipedes

Reptiles: Snakes, lizards, Turtles

Mammals: Humans, Rat, monkeys

② Characteristics

Arthropods: (i) They have segmented bodies

(ii) The members of this phylum all possess an exoskeleton

Reptiles: (i) They are poikilothermic i.e. they are cold blooded

(ii) The bodies of the members of this class are completely covered with scales.

Mammals: (i) Mammals are warm-blooded

(ii) They are heterodont, i.e., possess different types of teeth.

③ The alternative name for phylogenetic tree is The Evolutionary Tree

④ Each group of organisms stated above shows specialization of organs. The phylum arthropods is the least complex of them all, followed by reptiles, mammals are the most complex group of them all. The basic relationship that exists from bottom to top is called an evolutionary relationship.

EXPERIMENT VII: ELEMENT OF ECOLOGY AND HABITATS OF PLANT AND ANIMAL.

ACTIVITY:

Identify and write names of the specimens provided.

Using the table below, state the ecological effect/relationship between each of the specimens to maize plant and man.

Specimen	Maize	Man
Air	Maize obtains carbon dioxide from the air for photosynthesis and pollination	Man breathes oxygen - present in the air for aerobic respiration
Sunlight	Sunlight is essential to the maize plant in order to carry out - photosynthesis	Man uses the energy in sunlight for daily activities. Man also obtain vitamin D from sunlight.
Stones and sand & soil	Stones and sand provide - anchorage and support for the maize plant.	Stones and sand are used in construction of - shelter for man.
Water	Water is required for the metabolic process of the maize plant.	Water helps is also essential for homeostasis in man.
Fire	Fire destroys the maize plant as it burns away the cells of the plant	Fire is used as a source of heat energy to man. It also causes harm to man by destroying the - skin cells of them.

EXPERIMENT VIII: ECOLOGICAL ADAPTATION OF PLANT AND ANIMAL FORMS

ACTIVITY I: TERRESTRIAL PLANTS

- Examine, draw and name specimen A provided (10-12cm) Cashew plant
- State two adaptive features of specimen A
 - A well developed palisade tissue for water storage
 - A well developed endodermis which serve as a barrier that controls and limit salt penetration in plant body
- Why/how is the plant adapted to its habitat?
 - It is adapted to its environment in order to get more nutrient from the soil.
 - It is done through intense lignification which is provided by the activity of succinate dehydrogenase and can be stimulated by salinity. Therefore, this could confer mechanical resistance of cells to high osmotic pressure of soil solution.

ACTIVITY II: AQUATIC PLANTS

- Name specimen B
Water hyacinth
- List the adaptive features of specimen B to its environment
 - Mechanical tissue like ~~aerenchyma~~ collenchyma and sclerenchyma are poorly developed
 - Lots of air sacs {Aerenchyma} for gaseous exchange/buoyancy.
- Why/how is the specimen adaptive to aquatic life?

Water hyacinth is a hydrophyte. It lacks cuticles so that it can get rid of excess water easily. It also possess swollen petioles that keep the plant afloat. Water hyacinth has reduced vascular tissues and absorbing too much water to prevent it from becoming turgid and eventually bursting.

ACTIVITY III: TERRESTRIAL ANIMALS

SPECIMEN C - LIZARD

- Carefully draw a well labeled diagram 10 -12 cm long of specimen C
- What behavioral adaptation does specimen C display to suit environmental conditions?
- How does specimen C minimize water loss? Mention two ways
 - Specimen C {lizard} uses camouflage to escape predators and hunt for food, it also warms itself by absorbing heat from the sun and ground.
 - The first way in which specimen C minimizes water loss is through their skin - they possess dry skin covered in scales made of keratin that minimizes water loss.
 - They also possess very efficient kidneys that are specially adapted to concentrate the body's waste products into uric acids, once the wastes are collected and converted, the reptile is able to reabsorb most of the liquid used in the process,

ACTIVITY IV: AQUATIC ANIMALS

Observe specimen D carefully and answer the following questions:

Identify specimen D

Tilapia fish

What is its habitat?

Tilapia fish is found in an aquatic habitat. They are fresh water fish inhabiting shallow streams, ponds, rivers and lakes.

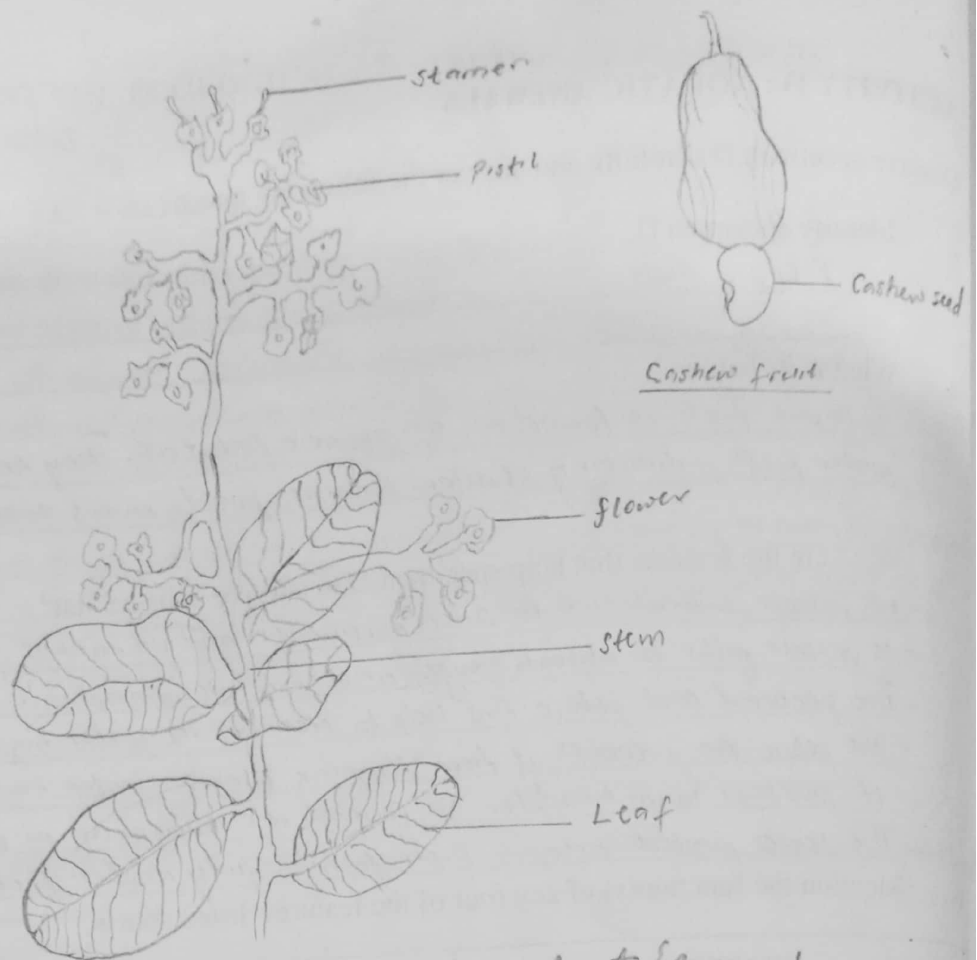
What are the features that help specimen D to survive in its habitat?

- It possess lateral line to sense movements and vibrations in the water.
- It possess gills to absorb oxygen and excrete ammonia.
- The pectoral and pelvic fins help to steer in the water.
- The pelvic fin prevents it from flipping over in water.
- It possess swim bladder to control its buoyancy in water.
- The scales protect tilapia fish from injury and infections

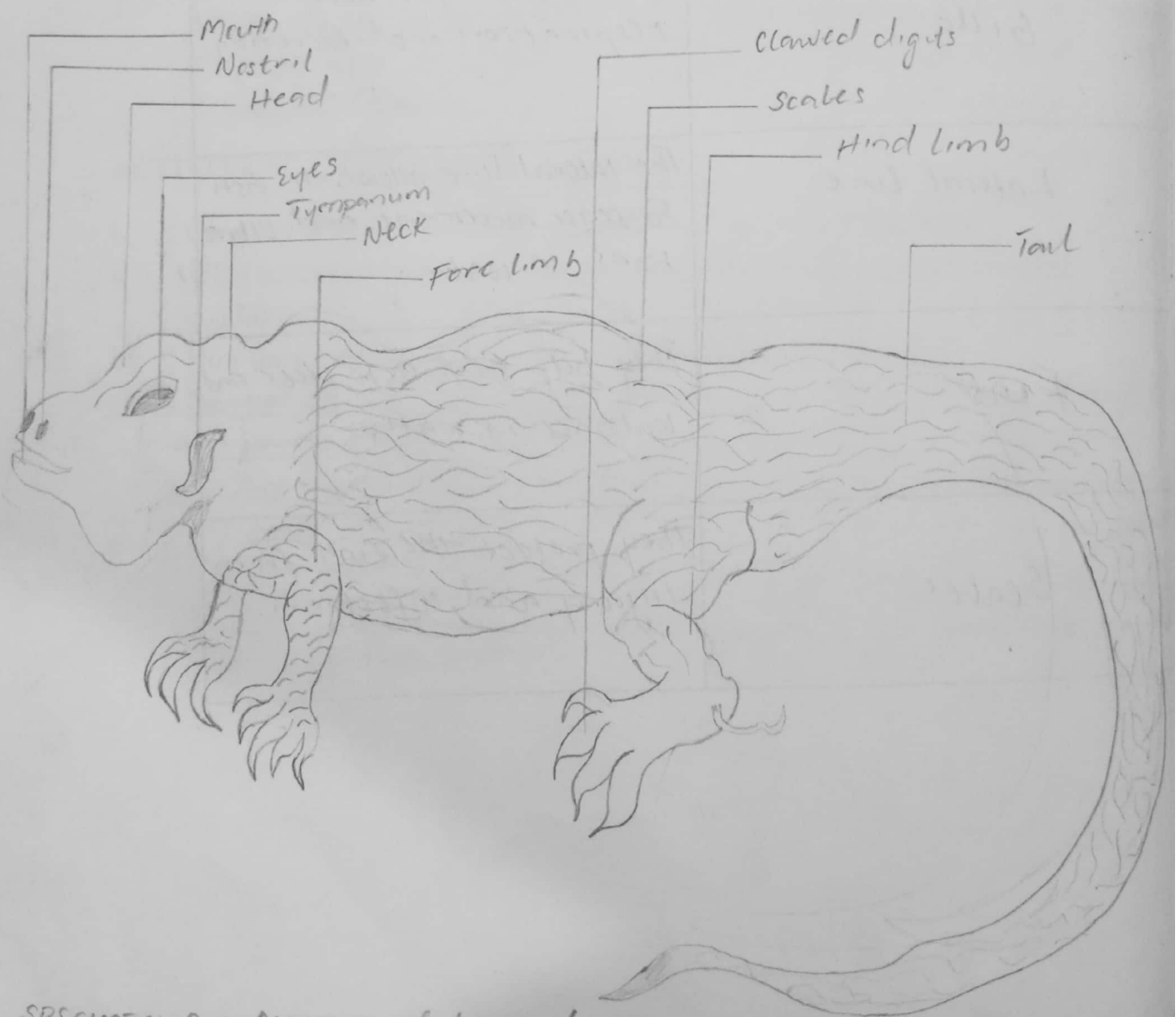
Mention the function(s) of any four of the features listed above

FEATURE	FUNCTION
Gills	It is used for both - respiration and excretion.
Lateral line	The lateral line allows the fish to sense movement and vibrations in water.
Fins	They help the fish to steer and balance in water.
Scales	They protect the fish from injury and infection.

17 January 2025



SPECIMEN A: Diagram of a cashew plant (*Anacardium occidentale*)



SPECIMEN C: Diagram of Lizard.